SECTIONS OF PULLED BACK MODULES ARE COLIMITS OF SECTIONS AT FINITE STAGES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1269-1292, following the statement at 1258-1267.

Editorial note. No omissions; the kernel exchange is the displayed final step.

Allowed standard prerequisites (editorial).

- Affine etale cover of a qcqs space. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover (cross-chapter, cited at ~1270-1272). Surjective etale U_0 -> X_0 affine.
- Presentation of the limit space. lemma-directed-inverse-system-has-limit (same chapter, card L26291, proof cited at ~1275-1277). X = U/R with U = lim U_i and R = lim R_i.
- Sections of pulled back modules on limits of schemes. Limits, Lemma limits-lemma-descend-section (cross-chapter, cited at ~1280-1286). F(U) = colim F_i(U_i) and F(R) = colim F_i(R_i).

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SI: focused-seeds/geometry/S9-spaces-limits-descend-section/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084R: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1250–1250. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H6: focused-seeds/geometry/S9-spaces-limits-descend-section/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Notation and assumptions as in Situation 084R. Suppose that $\mathcal{F}_0$ is a quasi-coherent sheaf on X_0. Set $\mathcal{F}_i = f_{0i}^*\mathcal{F}_0$ for $i \geq 0$ and set $\mathcal{F} = f_0^*\mathcal{F}_0$. Then*

$$\Gamma(X, \mathcal{F}) = \operatorname{colim}_{i \geq 0} \Gamma(X_i, \mathcal{F}_i)$$

Proof. Choose a surjective étale morphism $U_0 \to X_0$ where U_0 is an affine scheme (Properties of Spaces, Lemma 03H6). Set $U_i = X_i \times_{X_0} U_0$. Set $R_0 = U_0 \times_{X_0} U_0$ and $R_i = R_0 \times_{X_0} X_i$. In the proof of Lemma 07SF we have seen that there exists a presentation $X = U/R$ with $U = \lim U_i$ and $R = \lim R_i$. Note that U_i and U are affine and that R_i and R are quasi-compact and separated (as X_i is quasi-separated). Hence Limits, Lemma 07SI implies that

$$\mathcal{F}(U) = \operatorname{colim} \mathcal{F}_i(U_i) \quad \text{and} \quad \mathcal{F}(R) = \operatorname{colim} \mathcal{F}_i(R_i).$$

The lemma follows as $\Gamma(X, \mathcal{F}) = \operatorname{Ker}(\mathcal{F}(U) \to \mathcal{F}(R))$ and similarly $\Gamma(X_i, \mathcal{F}_i) = \operatorname{Ker}(\mathcal{F}_i(U_i) \to \mathcal{F}_i(R_i))$ □